

Project Support Session

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Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

Project Problem Statement

Business Context

As organizations grow, business analysts increasingly face the challenge of navigating large volumes of reports, research papers, and strategic documents. Extracting the right insights from lengthy materials can be time-consuming and overwhelming, especially when these insights directly influence key business decisions.

Consider joining a venture capital firm like Andreessen Horowitz and being assigned a dense report such as Harvard Business Review's "How Apple is Organized for Innovation." Manually reviewing such documents requires significant effort, slowing down the analysis process and increasing the chances of missing important details.

To overcome this information overload, businesses can leverage Semantic Search and Retrieval-Augmented Generation (RAG) models. These systems allow analysts to ask natural-language questions like, "How does Apple structure its teams for innovation?" and instantly retrieve relevant, accurate insights from the source document.

By integrating such AI-driven retrieval systems, organizations can streamline research workflows, reduce manual effort, and enable analysts to focus on high-value strategic thinking, ultimately improving decision-making speed and quality.

Project Problem Statement

Objective

As an AI specialist, your task is to develop a RAG-based application that enables business analysts to efficiently extract insights from extensive business reports such as “How Apple is Organized for Innovation.” The objective is to understand the challenges of navigating long, information-dense documents, apply retrieval-augmented generation techniques to surface only the most relevant content, analyze how this improves the speed and accuracy of report interpretation, evaluate its potential to enhance strategic decision-making and productivity for analysts, and create a functional prototype that demonstrates the system’s effectiveness in answering queries, summarizing key insights, and supporting natural-language interactions without requiring users to read the entire report.

Dataset Overview

- “How Apple is Organized for Innovation” is a detailed Harvard Business Review article that examines Apple’s unique approach to structuring teams, driving innovation, and maintaining a culture of excellence. The article offers in-depth insights into Apple’s organizational design, leadership principles, and decision-making processes.
- The article is a PDF with 11 pages.



Project Evaluation Rubric

Section	Points
Question Answering using LLM with Generic Prompts	10
Question Answering using LLM with Prompt Engineering	14
Data Preparation for RAG	12
Question Answering using RAG	12
Actionable Insights & Recommendations	4
Presentation/Notebook - Overall Quality	8

Project Evaluation Rubric

Question Answering
using LLM with
Generic Prompts

Question Answering
using LLM with
Prompt Engineering

Data Preparation
for RAG

Question Answering
using RAG

Actionable Insights &
Recommendations

Presentation /
Notebook Quality

What to Cover?

Load the large language model

Create a function to define the
model parameters and response

Guiding Questions and Considerations

Load any appropriate open-source LLM from Hugging Face or
proprietary models from OpenAI

Define a function that accepts a text query along with other parameters
needed for the LLM

Project Evaluation Rubric

Question Answering
using LLM with
Generic Prompts

Question Answering
using LLM with
Prompt Engineering

Data Preparation
for RAG

Question Answering
using RAG

Actionable Insights &
Recommendations

Presentation /
Notebook Quality

What to Cover?

Apply the response generation
function to get answers to the
questions provided in the problem
statement

Provide comments/observations for
the answers received

Guiding Questions and Considerations

Use the previously defined function on all the given questions to
generate the responses

How do the responses look like? Are they too generic?

Project Evaluation Rubric

Question Answering
using LLM with
Generic Prompts

Question Answering
using LLM with
Prompt Engineering

Data Preparation
for RAG

Question Answering
using RAG

Actionable Insights &
Recommendations

Presentation /
Notebook Quality

What to Cover?

Apply prompt engineering and LLM parameter tuning and get answers to the questions provided in the problem statement

Provide comments/observations for the answers received

Guiding Questions and Considerations

What factors to consider when engineering a better prompt for the desired output?

What LLM parameters can be tuned for creativity vs specificity?

How does the response change compared to a Baseline LLM?

Project Evaluation Rubric

Question Answering
using LLM with
Generic Prompts

Question Answering
using LLM with
Prompt Engineering

**Data Preparation
for RAG**

Question Answering
using RAG

Actionable Insights &
Recommendations

Presentation /
Notebook Quality

What to Cover?

Load the data file provided

Split the data using a text splitter
with necessary attributes

Load the embedding model

Guiding Questions and Considerations

Use the appropriate PDF loader to load the PDF data file.

What are the considerations for choosing chunk size and chunk overlap?

How to choose an embedding model?

Project Evaluation Rubric

Question Answering
using LLM with
Generic Prompts

Question Answering
using LLM with
Prompt Engineering

**Data Preparation
for RAG**

Question Answering
using RAG

Actionable Insights &
Recommendations

Presentation /
Notebook Quality

What to Cover?

Load the vector database

Define the retriever with
appropriate search method and k
value

Guiding Questions and Considerations

The vector database enable faster retrieval of relevant document(s)

Small values of k (e.g., 3–5) for precise, focused responses and faster
retrieval

Large values of k (e.g., 10–20) if the context is broad or the query
requires more supporting information

Project Evaluation Rubric

Question Answering
using LLM with
Generic Prompts

Question Answering
using LLM with
Prompt Engineering

Data Preparation
for RAG

Question Answering
using RAG

Actionable Insights &
Recommendations

Presentation /
Notebook Quality

What to Cover?

Get answers to the questions
provided in the problem statement

Fine-tune the chunking, retriever,
and LLM parameters to check
different results

Provide comments/observations for
the answers received

Guiding Questions and Considerations

Get the responses for all the questions provided

Try different combinations of RAG and LLM parameters

How does the response change on using RAG?

Project Evaluation Rubric

Question Answering
using LLM with
Generic Prompts

Question Answering
using LLM with
Prompt Engineering

Data Preparation
for RAG

Question Answering
using RAG

Actionable Insights &
Recommendations

Presentation /
Notebook Quality

What to Cover?

Conclude with the key takeaways
for the business

Guiding Questions and Considerations

What are the key things that we learned from the data that's useful for
the business?

Logical next steps are a key inclusion

Project Evaluation Rubric

Question Answering
using LLM with
Prompt Engineering

Data Preparation
for RAG

Question Answering
using RAG

Output Evaluation

Actionable Insights &
Recommendations

Presentation /
Notebook Quality

Low-Code Considerations

Clear structure, flow, and visual appeal - everything sits well in a story

Crispness - Focus on key points

All sections of the rubric included, even if in an appendix

Full-Code Considerations

Clear structure and flow

Well-commented and executed code

All sections of the rubric included, even if in an appendix

Project Submission Guidelines: Low-Code

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Learner Notebook - Low Code* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

Project Submission Guidelines: Low-Code

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

Project Submission Guidelines: Full-Code

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution from scratch

Download the dataset and the *Learner Notebook - Full Code* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

Project Submission Guidelines: Full-Code

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via [free online tools](#)

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

1. The vector database is taking longer than expected to be created. How to handle it?

Chunk size and chunk overlap control the time and space complexity of the vector database.

Lowering the chunk size will help reduce the memory needed to store these chunks.

Lowering the chunk_overlap parameter will reduce the number of generated chunks.

2. I am getting the below error. How do I resolve it?

▼ Installing Necessary Libraries and Dependencies

```
[1] # Installation for GPU llama-cpp-python
!CMAKE_ARGS="-DLLAMA_CUBLAS=on" FORCE_CMAKE=1 pip install llama-cpp-python --force-reinstall --upgrade --no-cache-dir -q

      8.8/8.8 MB 21.8 MB/s eta 0:00:00
Installing build dependencies ... done
Getting requirements to build wheel ... done
Installing backend dependencies ... done
Preparing metadata (pyproject.toml) ... done
      18.2/18.2 MB 148.7 MB/s eta 0:00:00
      45.5/45.5 kB 232.9 MB/s eta 0:00:00
Building wheel for llama-cpp-python (pyproject.toml) ... done
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
lida 0.0.10 requires fastapi, which is not installed.
lida 0.0.10 requires kaleido, which is not installed.
lida 0.0.10 requires python-multipart, which is not installed.
lida 0.0.10 requires uvicorn, which is not installed.
llmx 0.0.15a0 requires cohere, which is not installed.
llmx 0.0.15a0 requires openai, which is not installed.
llmx 0.0.15a0 requires tiktoken, which is not installed.
tensorflow-probability 0.22.0 requires typing-extensions<4.6.0, but you have typing-extensions 4.9.0 which is incompatible.
```

2. I am getting the below error. How do I resolve it?

Kindly ensure that the runtime is connected to a T4 GPU, as DLLAMA_CUBLAS is set to 'on'.

Please note that this is not an error but a dependency conflict issue. It can be safely ignored, as the required versions of the packages have been successfully installed and will work smoothly.

Kindly restart the session after the installation command is executed to ensure that the environment is updated correctly.



Power Ahead!

